GBDA 302 – Global Digital Project 2
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Side Quest Week 6 - Process & Decision Documentation

1. What I was trying to do

For Week 6, my goal was to expand the platformer by improving the visual presentation and adding audio feedback. The focus of this side quest was integrating custom sprite sheets, replacing the default environment graphics with a tileset, and implementing sound effects that respond to gameplay events.

The intention was to make the game feel more complete and responsive by connecting player actions with animations and sounds. Instead of using placeholder graphics, the player character, enemies, and collectibles now use sprite sheets that animate as the game runs. Additionally, sound effects were added to reinforce important interactions such as jumping, attacking, collecting items, and taking damage.

By combining these elements, the game becomes more immersive and visually expressive while still maintaining the modular structure established in previous weeks.

2. One small decision or change I made

One important decision was to integrate sound using the event-driven system already present in the project. Instead of directly playing sounds inside gameplay logic, I connected sound effects to gameplay events such as player:jumped, player:attacked, and leaf:collected.

This approach keeps the code organized by separating gameplay logic from audio playback. The player controller emits events when actions occur, the game system listens for those events, and the sound manager handles playing the appropriate sound.

Another decision was to replace the default graphics with a consistent tileset and sprite sheets. This required adjusting sprite loading, animation configuration, and frame alignment so that the new graphics worked correctly with the existing animation system.

3. Evidence of the change

- Several updates were implemented to demonstrate the integration of sprites, tiles, and sound.
- Custom sprite sheets were added for the player character, enemies, and collectibles. These sprites are loaded during the asset loading stage and animated through configuration defined in the tuning JSON file. This allows different animations such as idle, running, jumping, attacking, and damage states to be displayed during gameplay.
- The environment graphics were also replaced with a tileset. Ground tiles, platform tiles, and wall tiles are now rendered using the custom tileset instead of placeholder graphics, giving the level a more cohesive visual style.
- Sound effects were added for several gameplay actions. These include jumping, attacking, collecting leaves, and receiving damage. The sound system uses an event-based architecture where gameplay systems emit events and the sound manager plays the corresponding audio file.
- In addition to sound effects, a looping background music track was added to play during gameplay. This creates a more immersive atmosphere and makes the game feel more complete.

Examples of implemented changes include:

- Player sprite animations replacing placeholder graphics

- Custom tileset used to render ground and platform elements
- Sound effects triggered when jumping, attacking, collecting items, and taking damage
- Looping background music during gameplay
- Event-driven sound system connecting gameplay events to audio playback

4. GenAI Use

I used GenAI (ChatGPT) to help understand the structure of the provided starter code and to troubleshoot issues that arose during development. In particular, GenAI assisted with debugging sprite sheet loading, configuring animation frames, and integrating sound effects into the event-driven architecture of the game. GenAI was also helpful in explaining how different modules in the project communicate with each other, including how the player controller emits events, how the game system listens for those events, and how the sound manager plays the corresponding audio files. All suggestions generated by AI were reviewed, tested, and adapted to ensure they worked correctly within the existing codebase.

Appendix: GenAI Transcript and Screenshots

Tool Used: ChatGPT

The full GenAI transcripts are extremely long; therefore, direct links are provided below for reference rather than embedding the transcripts in this document.

https://chatgpt.com/share/69adab35-c918-8009-ae14-e6703ed49325

https://chatgpt.com/share/69adab71-b8f4-8009-9288-ea6cb51b3f90

Thought for 38s

Fix these four spots.

1. In `preload()`, change the bad tile filenames.

Right now your uploaded `sketch.js` loads:

```javascript
groundImg   = loadImage("assets/tile_ground.png");
platformImg = loadImage("assets/tile_platform.png");
```

Those are the wrong names for the assets you showed earlier. This is at `preload()` in `sketch.js`. sketch

Replace that block with:

```javascript
function preload() {
  playerImg = loadImage("assets/foxSpriteSheet.png");
  enemyImg = loadImage("assets/Boar_Attack_with_shadow.png");
  groundImg = loadImage("assets/groundTile.png");
  platformImg = loadImage("assets/platformLC.png");
}
```

2. In `buildLevel()`, keep `t.image = ...` but know what it will show.

You are drawing every platform tile with the same image:

```javascript
t.image = platformImg;
```

That means every platform block will use only `platformLC.png`, which is a left-cap tile. So even after fixing the file path, the platforms may look odd. That code is in `buildLevel()`. sketch

For now, leave it if you just want it working.
Later, if you want it to look better, use a middle tile instead of `platformLC.png`, or load both left and right cap images and place them separately.

3. In `spawnPlayer()`, do not change anything yet.

Your fox sheet is still set up correctly for 32×32 frames:

```javascript
player = new Sprite(48, PLAYER_Y, 32, 32, "d");
player.anis.w = 32;
player.anis.h = 32;
player.addAnis(foxAnis);
```

That part is fine if you are still using `foxSpriteSheet.png`. sketch

4. In `spawnEnemy()`, do not change anything yet either.

Your boar setup is also using 32×32 frames, which matches the comment in your code for that boar sheet. sketch sketch

So this part can stay:

```javascript
enemy = new Sprite(220, PLAYER_Y, 32, 32, "d");
enemy.anis.w = 32;
enemy.anis.h = 32;
enemy.addAnis(boarAnis);
```

README.md
File

sketch.js
JavaScript

jsconfig.json
File

index.html
File

style.css
File

i am glithcing out when i added new wlemenst what wrong

Thought for 1m 22s

The most likely problem is your images are inconsistent, not the leaf logic.

In your code, the leaves are not image assets at all. They are drawn manually as green ellipses and a stem in `drawLeaves()`, which is why they still appear even when other game visuals are broken. Your player, enemy, and tiles depend on loaded image files in `preload()`. sketch

The biggest red flag is this part:

```javascript
groundImg   = loadImage("assets/tile_ground.png");
platformImg = loadImage("assets/tile_platform.png");
```

But earlier your assets folder showed names like `groundTile.png`, `groundTileDeep.png`, `platformLC.png`, `platformRC.png`, and `oakwoodtileset.png`, not `tile_ground.png` or `tile_platform.png`. That mismatch would explain why the level visuals disappear and the scene looks broken. sketch

So fix the filenames first. For example:

```javascript
function preload() {
  playerImg   = loadImage("assets/foxSpriteSheet.png");
  enemyImg    = loadImage("assets/Boar_Attack_with_shadow.png");
  groundImg   = loadImage("assets/groundTile.png");
  platformImg = loadImage("assets/platformLC.png");
}
```

That is the first thing to test.

Second, your README and code are out of sync. The README says the tiles are in `assets/tileset/` and mentions different tile names, while the actual sketch loads different files directly from `assets/`. That means the project structure is inconsistent right now.

Third, if the player and boar still do not show after fixing filenames, add these temporary debug lines at the end of `draw()`:

```javascript
fill("red");
circle(player.x, player.y, 8);

if (enemy) {
  fill("yellow");
  circle(enemy.x, enemy.y, 8);
}
```

If you see the red and yellow dots but no sprites, the sprite sheets are the issue. If you do not see the

EXPLORER

JMANJOOR_SIDEQUEST...
- assets
- data
  - highscores.json
  - levels.json
  - tuning.json
- libraries
  - p5.js
  - p5.sound.min.js
  - p5play.min.js
  - planck.min.js
  - q5.min.js
- src
  - entities
    - Boar.js
    - Fire.js
    - Leaf.js
    - PlayerEntity.js
  - ui
    - LoseScreen.js
    - WinScreen.js
  - utils
    - IntegerScale.js
  - world
  - AssetLoader.js
  - CameraController.js
  - DebugOverlay.js
  - EventBus.js
  - Game.js
  - HighScoreManager.js
  - InputManager.js
  - Level.js
  - LevelLoader.js
  - ParallaxBackground...
  - SoundManager.js
- types / p5
  - global.d.ts
  - index.d.ts
- .gitattributes
- index.html

OUTLINE

TIMELINE

levels.json | AssetLoader.js | main.js | README.md | tuning.json | Preview README.md | Game.js | SoundMana

data > levels.json > [ ] levels > { } 0 > [ ] tilemap > abc 12

if (t

You, 17 hours ago | 1 author (You)

```
{
  "schemaVersion": 1,
  "levels": [
    {
      "id": "ex5_level1",
      "name": "Example 5 Level",

      "tiles": {
        "tileW": 24,
        "tileH": 24,
        "frameW": 32,
        "frameH": 32
      },

      "tilemap": [
        "                    g   g   b  x        ",
        "                b x         LggR        ",
        "      x   f     LggR                    ",
        "     LR   LgR          LR               ",
        "   fx  b        x   b                   ",
        "   LgggR   x   LR   LgR x   b  xf       ",
        "          LgR  b x       g   LggggR     ",
        " fx           LgR                   fx ",
        " LgR      b                        LggR",
        "         LgR        f x    LR  LgR  [dd]",
        "   x     [d]      x LggR   x    ff  [dd]",
        "LgggRffLgggggggRfffLgggg]fffgfLgggggggggg",
        "dddddddddddddddddddddddddddddddddddddddd"
      ],

      "rules": {},

      "winScore": 15,
      "gravity": 10,
      "fallResetMarginTiles": 3,

      "playerStart": { "x": 32, "y": 216 },

      "view": {
        "w": 240,
        "h": 192,
        "background": [69, 61, 79],
        "cameraLerp": 0.1,

        "parallax": [
          { "img": "assets/bg3.png", "speed": 0.2 },
          { "img": "assets/bg2.png", "speed": 0.4 }
```

You, 21 hours ago • commit